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organisms are classified into prokaryote, eukaryote and mesokaryote.

The eukaryotic cells originated by endosymbiosis of prokaryotic organism. Key difference between plant cell and animal cell is the cell wall. Protoplasm is the colourless mass includes the cytoplasm, cell organelles and nucleus. Cell wall is the outermost protective covering with three regions primary, secondary wall and middle lamellae. Cell membrane holds the cytoplasmic content called **cytosol**. Cytoplasm includes the matrix and the cell organelles excluding nucleus. Endomembrane system includes endoplasmic reticulum, golgi apparatus, chloroplast, lysosomes, vacuoles, nuclear membrane and plasma membrane. Nucleus is the control unit of the cell, it carries hereditary information. Chromosomes are made up of DNA and associated proteins. Bacterial flagella are made up of helical polymers of a protein called **flagellin**. Proton motive force are involved in flagellar rotation. In Eukaryotes flagella are made up microtubules and protein called **dynein** and **nexin** and the movement is driven by ATP. Cytological techniques include preparation of slides, staining and recording the observation.

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Evaluation

1. The two subunits of ribosomes remain united at critical ion level of
   a. Magnesium b. Calcium
   c. Sodium d. Ferrous
2. Sequences of which of the following is used to know the phylogeny
   a. mRNA b. rRNA
   c. tRNA d. Hn RNA
3. Many cells function properly and divide mitotically even though they do not have
   a. Plasma membrane b. cytoskeleton
   c. mitochondria d. Plastids
4. Keeping in view the fluid mosaic model for the structure of cell membrane, which one of the following statements is correct with respect to the movement of lipids and proteins from one lipid monolayer to the other
   a. Neither lipid nor proteins can flip-flop
   b. Both lipid and proteins can flip flop
   c. While lipids can rarely flip-flop proteins cannot
   d. While proteins can flip-flop lipids cannot
5. Match the columns and identify the correct option:

Column-I	Column-II
(a) Thylakoids	(i) Disc-shaped sacs in Golgi apparatus
(b) Cristae	(ii) Condensed structure of DNA
(c) Cisternae	(iii) Flat membranous sacs in stroma
(d) Chromatin	(iv) Infoldings in mitochondria

	(a)	(b)	(c)	(d)
(1)	(iii)	(iv)	(ii)	(i)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iii)	(i)	(iv)	(ii)

6. Bring out the significance of phase contrast microscopy
7. State the protoplasm theory
8. Distinguish between prokaryotes and eukaryotes
9. Difference between plant and animal cell
10. Draw the ultra structure of plant cell